Oriented sources for the charged memory field

Exact transfer, whole-time bounds, and a conserved charge background

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Abstract

The actual adjacent-charge transfer defect admits an exact full-Fock loss-generator representation. We prove its weighted, compact-time replacement by a Gaussian-number/body-loss field and give summable ordinary-power tails. A source-dependent local normalization then computes fixed-degree common-coupling moments, including contractions internal to each field. Such contractions produce a conserved charge background: even diagrams with no direct Gaussian bridge between the fields can have a nonzero connected coefficient. We identify this coefficient and independently verify its lowest-degree realization. These results supply the charged input to the previously constructed neutral kernel; the complete charged static subtraction and joint dynamic kernel are left for the next stage.

1 Setting and the exact adjacent-charge identity

Use standard CAR, overlapping complex Gaussian p-body interactions, the same original couplings J across every charge sector, and normalized annealed full-Fock trace at $\beta = \mu = 0$. Take even N, $p \to \infty$, $\lambda_N = p^2/N \to \lambda > 0$, fixed J_s, $\hbar = 1$, and string time. Write $C = J_s^2/\lambda$, $q = e^{-4\lambda}$, and $b_{N,m} = 2^{-N}\binom{N}{m}$. Polynomial degrees and real times are fixed before this limit. Gaussian number $\mathcal{N}$ refers to coupling chaos, not spectral q-Hermite degree.

On neutral full-Fock operators define

$$DX = QX - \sum_i c_i^\dagger X c_i, \qquad Q = \sum_i c_i^\dagger c_i. \tag{1}$$

For $m > 0$, $h = H|_m$ and $T_m X = \sum c_i^\dagger X c_i/m$, hence

$$F_m(s) = T_m e^{isH_{m-1}} = e^{ish} - [De^{isH}]_m/m, \tag{2}$$

$$E_m(s) = m(F_m - \alpha_m^{\mathcal{N}} e^{ish}) = [m(1-\alpha_m^{\mathcal{N}}) - D]e^{isH}|_m, \tag{3}$$

$$\alpha_m = (1 - p/m)_+. \tag{4}$$

We set $E_0 = 0$; $m < p$ is inactive and $E_m = 0$. At $m = p$ the convention $0^0 = 1$ preserves the constant chaos. Since $DI = 0$ and $DH = pH$, $E(0) = E'(0) = 0$.

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2 Weighted replacement for the entire exponential

Retain the finite parameter λ_N and put

$$G_N(s) = [p\mathcal{N} - D - \lambda_N \mathcal{N}(\mathcal{N}-1)]e^{isH}. \tag{5}$$

All its restrictions to $m < p$, including $m = 0$, vanish. For Gaussian matrix-valued functions use $\|\psi\|_r = \|(\mathcal{N}+1)^{r/2}\psi\|_2$, summed with $b_{N,m}$. This is a Gaussian Sobolev norm. The preceding weighted-transfer estimates [3] give, sectorwise with $C_m \le M$,

$$\|H_{L/R}\psi\|_r \le A_r\sqrt{M}\|\psi\|_{r+1}, \qquad \|e^{itH_L}\psi\|_r \le e^{q_r\sqrt{M}|t|}\|\psi\|_r, \tag{6}$$

where $A_r = 1 + 2^{r/2}$ and $q_r = r2^{\max(r-1,0)}$.

Proposition 1. *For fixed $r, j, a, b \ge 0$, $S < \infty$, and λ_N in a positive compact interval,*

$$\sup_{|s|\le S} \|H^a \partial_s^j (E - G_N)(s) H^b\|_r = O(p^{-1}). \tag{7}$$

Proof. The D terms cancel exactly. For $k \ge 0$ and $u = p/m \le 1$, Taylor's integral remainder gives

$$0 \le m[1-(1-u)^k] - kp + \binom{k}{2}p^2/m \le \binom{k}{3}p^3/m^2. \tag{8}$$

For $k < 3$ the right side is zero. On $N/4 \le m \le 3N/4$ the remaining number multiplier $R_m(k) = m(1-\alpha_m^k) - kp + \lambda_N k(k-1)$ obeys

$$|R_m(k)| \le \left(\frac{4\lambda_+|m-N/2|}{N} + \frac{8\lambda_+^2}{3p}\right)(k+1)^3. \tag{9}$$

Apply the Sobolev estimates to $H^j e^{isH}$, and then the left/right energy multipliers. The binomial identity $\sum_m b_{N,m}(m-N/2)^2 = N/4$ gives the squared error $O(p^{-2})$. On the active complement, $|R_m(k)| \le (p+\lambda_+)(k+1)^2$; its binomial weight is at most $2e^{-N/8}$. These estimates also give the stated uniformity in s. □

The force interface is exact:

$$\mathcal{Q}_\alpha f = -f'' + i\alpha\{h, f'\} + \alpha^2 h f h, \tag{10}$$

$$m(\Lambda - \Lambda_{\mathrm{OU}}) = \mathcal{Q}_{\alpha_m} E_m, \tag{11}$$

$$\mathsf{A}_m(s) = (1-\mathcal{P}_H)e^{-ihs/2}(\mathcal{Q}_{\alpha_m}E_m)(s)e^{-ihs/2}. \tag{12}$$

Here $\mathcal{P}_H X = \sum_e P_e X P_e$ retains each full samplewise energy eigenspace, including degeneracies, within the relevant charge sector. Weighted transfer bounds E and G_N; $\alpha_m - 1 = O(1/p)$ on the broad central window. Thus $\mathcal{Q}_{\alpha_m}E_m$ may be replaced by $\mathcal{Q}_1 G_N$ in full-trace L^2. Replacing λ_N by λ would require the separate error $O(|\lambda_N - \lambda|)$.

3 An ordinary-power expansion that retains lower chaos

Define the complete coefficients

$$G_n = [p\mathcal{N} - D - \lambda_N \mathcal{N}(\mathcal{N}-1)]H^n, \qquad G_N(s) = \sum_{n\geq 0} \frac{(is)^n}{n!} G_n. \tag{13}$$

Write $E_{n,m} = m[T_m H^n_{m-1} - \alpha^{\mathcal{N}}_m h^n]$ for the corresponding exact transfer coefficients, with the preceding inactive sector conventions. Here $G_0 = G_1 = 0$. We do not replace this series by its highest-chaos terms. In particular, D differentiates charge-dependent contraction coefficients.

The polar map $T = \mathcal{U}S_- = S_+\mathcal{U}$ from [2] has a Gaussian-field defect bounded on chaos k by $K\sqrt{MB_m}(k+1)^{5/2}/N$, where

$$B_m = \lambda_N + \lambda_N^2 + \lambda_N(2m - N - 1)^2/N. \tag{14}$$

Telescoping ordinary products and using the Gaussian creation/annihilation bound, with $L = 2\sqrt{M}$, gives

$$\|\mathcal{U}H^n_- - H^n\|_2 \leq \frac{K\sqrt{MB_m}}{N} L^{n-1}\sqrt{n!} \sum_{a<n} (a+1)^2. \tag{15}$$

The rank-filter estimate for $S_+ - \alpha^{\mathcal{N}}$ is applied to the whole H^n. Multiplication by $m \leq N$, followed by binomial summation, therefore yields

$$\|E_n\|_2 + \|G_n\|_2 \leq K(n+1)^3 L^n \sqrt{n!}. \tag{16}$$

The difference $E_n - G_n = R(\mathcal{N})H^n$ obeys the same type of bound by (9). Outside the central window, polynomial factors in N, p are absorbed by $e^{-N/8}$. Fixed Sobolev orders, derivatives, or left/right energy powers only add a fixed power of $n+1$.

Consequently the force coefficients are

$$K_n = G_{n+2} - \{H, G_{n+1}\} + HG_nH, \tag{17}$$

$$(\mathcal{Q}_1 G_N)(s) = \sum_{n\geq 0} \frac{(is)^n}{n!} K_n, \tag{18}$$

$$\|K_n\|_2 \leq K'(n+3)^4 L^{n+2}\sqrt{n!}. \tag{19}$$

For every fixed S the corresponding tails are bounded by sums of $\mathrm{poly}(n)(LS)^n/\sqrt{n!}$, which converge. The same tails hold after exact energy pinching, a Hilbert-space contraction. They allow a fixed-degree spectral construction to be extended to finite memory times once its static sector has been identified.

The small-time calibration is also transparent: $\mathsf{A}(0)$ is the pinched G_2. The preceding results require this to reproduce Y_-; higher coefficients must reproduce the nonzero mixed spectra in [4]. Bounds alone do not establish those spectral identities.

4 Oriented marks on pure coupling-Wick sources

The neutral size operator is $\mathcal{L}X = NX/2 - \frac{1}{2}\sum_i (c_i^\dagger X c_i + c_i X c_i^\dagger)$, half the Clifford length, as in the preceding neutral construction. For a globally neutral even source, the loss generator may be written as a sum of symmetric local generators:

$$D_i x = \tfrac{1}{2}\{n_i, x\} - c_i^\dagger x c_i, \qquad x = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad D_i x = \begin{pmatrix} 0 & b/2 \\ c/2 & d-a \end{pmatrix}. \tag{20}$$

The asymmetric table is $n_i x - c_i^\dagger x c_i$; its sum agrees on $[Q, X] = 0$. The asymmetric deficit uses the raw creation count n_c, whereas the symmetric one uses $(n_c + n_a)/2$. Their equality follows from $[n_i, x] = (n_c - n_a)x$. All fermion reordering signs must be retained before local traces.

For a j-slot source with no internal Gaussian pairing, let I be its incident chords. The one-mark exponents for $jp - D$, $jp - \mathcal{L}$, and $jp - \bar{D}$, with $\bar{D} = 2\mathcal{L} - D$, are

$$A_D = j(j-1) - 4\,\mathrm{cr}(I), \quad A_L = j^2 - 4\,\mathrm{cr}(I), \quad A_{\bar{D}} = j(j+1) - 4\,\mathrm{cr}(I). \tag{21}$$

If two such marked sources share b chords with c mutual crossings, the connected exponent K_{12} is

Pair	K_{12}
DD	$2b(b-1) - 4c$
DL or LD	$b(b-1) - 4c$
LL	$b^2 - 4c$
$D\bar{D}$	$-4c$
$L\bar{D}$	$b(b+1) - 4c$
$\bar{D}\bar{D}$	$2b(b+1) - 4c$

The double-mark weight of a complete pairing is

$$C^{\#\text{pairs}} q^{\mathrm{cr}\,\pi} [\lambda K_{12} + \lambda^2 A_1 A_2]. \tag{22}$$

The one-chord D deficit is identically zero, as required by $DH = pH$. Negative mixed terms and zero unmarked traces with nonzero marked coefficients are retained. The pure-source regression coefficients obey $\kappa_j^D = \kappa_j^L - \lambda j$ and $\kappa_j^{\bar{D}} = \kappa_j^L + \lambda j$.

These formulas are derived here using the finite-CAR and pair-Poisson framework of [1, Sec. 3.2–3.3]; they are not quoted as formulas of that paper. A nonadjacent two-slot determinant shortcut fails if the whole source permits internal contractions. The next construction handles exactly that case.

5 Whole-field sources and singleton normalization

Fix a complete Gaussian pairing π of 2ℓ ordered slots. There are 2ℓ independent p-set labels; label a supplies the creator at slot a and the annihilator at slot $\pi(a)$. For a marked whole-field interval F let

$$i_F = \#\{\text{pairs internal to } F\}, \qquad d_F = |F| - 2i_F. \tag{23}$$

Gaussian Wick decomposition selects degree d_F, including its internal contractions. Introduce commuting nilpotents $z_F^2 = 0$. The source is $\exp[z_F(pd_F - D_F)]$. The local semigroup follows from (20):

$$e^{-zD_i} c_i^{(\dagger)} = e^{-z/2} c_i^{(\dagger)}, \qquad e^{-zD_i} n_i = e^{-z} n_i, \tag{24}$$

$$e^{-zD_i}(1 - n_i) = 1 - e^{-z} n_i. \tag{25}$$

Cancel singleton factors before taking the limit. A chord open at F gives two labels, each with factor $e^{-z_F/2}$ at each of its p sites. This cancels $e^{pd_F z_F}$. An internal chord gives the normalized singleton factors e^{-z_F} and $2 - e^{-z_F}$, whose product is 1 in the nilpotent ring. This cancellation is exact, not an expansion leaving an $O(p)$ error. With several disjoint fields it holds in each source variable.

Let $t_a(z)$ denote the full local trace when only label a occupies a site, and $t_{ab}(z)$ the merged-site trace. Then $t_a(0) = 1/2$ and the ratio for a collision is

$$R_{ab}(z) = \epsilon_{ab} \frac{t_{ab}(z)}{t_a(z) t_b(z)}, \qquad \mathcal{K}_\pi(z) = \sum_{a<b} [R_{ab}(z) - 1] - \ell. \tag{26}$$

Here ϵ_{ab} is the exact fermion-grouping sign. Division is only by singleton polynomials with nonzero constant term, never by a possibly zero merged trace. The $-\ell$ fixes the unmarked variance normalization. One obtains $\mathcal{K}_\pi(0) = -4 \operatorname{cr} \pi$.

Pair intersections converge jointly to independent Poisson variables of mean λ. Triple intersections vanish; fixed-degree source coefficients are dominated by a polynomial times an exponential of the total overlap count, whose moments are uniformly bounded. Thus the diagram weight is $C^\ell \exp(\lambda \mathcal{K}_\pi)$. For G insertions shift each one-source coefficient by $-d_F(d_F - 1)$; the connected coefficients are unchanged. For two G fields this gives the explicit rule

$$C^\ell q^{\operatorname{cr} \pi} \left[\lambda \kappa_{12} + \lambda^2 (\kappa_1 - d_1(d_1 - 1))(\kappa_2 - d_2(d_2 - 1)) \right]. \tag{27}$$

External energy slots are excluded from d_F. Mixed neutral marks use their own Wick-source restrictions and the local $\mathcal{L}$ table.

6 A closed geometric rule for all ordinary powers

The local construction admits a useful simplification. For field a, let O_a be the chords with exactly one endpoint in that field, and let i_a count its internal chords. Thus $d_a = |O_a|$. For two fields put $B = O_1 \cap O_2$, the direct bridge chords, and $b = |B|$. Then

$$\kappa_a = d_a(d_a - 1) - 4\,\mathrm{cr}(O_a) - 2i_a, \tag{28}$$

$$\kappa_{12}^{DD} = 4i_1 i_2 + 2b(b-1) - 4\,\mathrm{cr}(B). \tag{29}$$

In particular the one-source exponent for G is simply $-4\,\mathrm{cr}(O_a) - 2i_a$. The G shift does not change the connected exponent. For a neutral Wick subblock let I_L be the chords incident to that marked subblock, and $B_{DL} = O_D \cap I_L$. Then

$$\kappa_{12}^{DL} = |B_{DL}|(|B_{DL}| - 1) - 4\,\mathrm{cr}(B_{DL}). \tag{30}$$

Its own single-source exponent is $|I_L|^2 - 4\,\mathrm{cr}(I_L)$. The set I_L may include chords internal to the larger neutral field; it equals that field's open set only when the field itself is pure Wick.

For the single-source identity, the two labels of an internal chord give -2, whereas those of an open chord give zero. Two open incident chords give the pure-source oriented rule. For two internal chords, or one internal and one open chord, the four possible label collisions have zero total single-source coefficient: crossing configurations have the relevant local zero word, and the nonzero noncrossing contributions cancel with opposite signs. The same cancellation holds for an internal chord and a spectator chord. An open leg and a spectator cancel against the singleton denominator. Thus spectator pairings contribute no extra derivative of the base q weight. For two sources, the internal–internal contribution is $4i_1 i_2$, and the remaining connected part comes from shared bridge chords, where the pure-source table applies.

This is a fixed-degree common-J moment rule, with arbitrary external energy insertions. It retains all lower-chaos contractions. Ordered energy multipliers on G_n do not belong to its source interval. Likewise, a neutral descendant has a whole Wick-source restriction and possibly a smaller marked subblock; these intervals must remain distinct, as in the preceding neutral construction [5].

Lowest dynamic calibration. Let $H_2^{(q,C)}(H) = H^2 - C$. The three diagrams for $\|G_2\|_2^2$ give

$$\lim \|G_2\|_2^2/C^2 = 8\lambda + 4\lambda^2 + 16\lambda^2 q, \tag{31}$$

$$g_{2,\mathrm{energy}}(H) = -2\lambda C - \frac{4\lambda q}{1+q} H_2^{(q,C)}(H). \tag{32}$$

After subtracting this deterministic energy regression, the squared norm is $C^2[8\lambda + 16\lambda^2 q/(1+q)]$. Removing the conserved charge residual identified on the next page subtracts another $4\lambda C^2$. The result is

$$w = 4C^2\left[\lambda + \frac{4\lambda^2 q}{1+q}\right] = 4J_s^4\left[\frac{1}{\lambda} + \frac{4q}{1+q}\right], \tag{33}$$

exactly the preceding Y_- variance. This calibrates the norm; it does not by itself prove the full two-energy measure or general pinching statement.

7 No direct field bridge does not mean no correlation

Suppose two whole-field D sources have no Gaussian chord directly connecting their intervals. Their connected exponent nevertheless is

$$\boxed{\kappa_{12} = 4i_1 i_2.} \tag{34}$$

To see this, a two-label collision can touch both fields only through internal–internal, internal–open, or open–open chords. An open leg scales by $e^{-z/2}$, which cancels against its source-dependent singleton denominator; hence the latter two cases have zero mixed coefficient. For each pair of internal chords, the four possible label collisions have mixed coefficients $0, 4, 0, 0$. Their sum is 4. This local classification is unaffected by spectator endpoints. It also shows that a pure neutral Wick subblock has no such mixed connected contribution in a diagram without a field bridge.

The smallest example has two H^2 fields, each paired internally. It gives

$$\mathcal{K} = -2z_1 - 2z_2 + 4z_1 z_2, \qquad [z_1 z_2] e^{\lambda \mathcal{K}} = 4\lambda + 4\lambda^2. \tag{35}$$

Subtracting only the averaged scalar leaves the positive term $4\lambda C^2$. It is a conserved charge fluctuation, not a dynamic correlation.

Indeed $\mathcal{W}_0 H^2 = C_Q I$ and the zero-chaos part of G_2 is exactly $-DC_Q$. For the common sector variance $C_m = w_N \binom{m}{p}\binom{N-m+p}{p}$, where $w_N = 4^p (J_s^2/\lambda_N)(p!/N^p)^2$, direct algebra gives

$$(DC_Q)_m = m(C_m - C_{m-1}) = C_m \frac{p(N - 2m + p + 1)}{N - m + 1}. \tag{36}$$

The formula includes $m = p$; inactive sectors are zero. In the central window $m = N/2 + xp$,

$$-DC_Q = -2\lambda C(1 - 2x) + o(1). \tag{37}$$

Under $b_{N,m}$, $X_N = (Q - N/2)/p$ tends to a centered Gaussian with variance $1/(4\lambda)$. A uniform central linear bound and the binomial tail give convergence of the first two moments:

$$\mathbb{E}\,\mathrm{tr}(-DC_Q) \to -2\lambda C, \quad \mathbb{E}\,\mathrm{tr}(DC_Q)^2 \to 4C^2(\lambda + \lambda^2), \quad \mathrm{Var}(-DC_Q) \to 4\lambda C^2. \tag{38}$$

This independently reproduces (35). Since $[Q, H] = 0$, exact energy pinching removes this background samplewise, including degenerate energy blocks. The general charged field still requires a joint charge/energy regression before the neutral no-bridge argument can be extended.

8 Checks, provenance, and the next calculation

Five new scientific outputs are exclusive-create; no predecessor driver was rerun. Their input and output hashes, dependency versions, parameters, build logs, and final rendered pages are recorded in the stage evidence file.

- Complete finite transfer: four CAR systems, 116 neutral matrix units, 4368 Gaussian-Hermite coefficient cases, 46 independent conditional-Gaussian OU polynomials, and 12582 exact Taylor cases. There are 17465 matrix equalities across 468408 integer components. The finite normalization is $C_m = 2\binom{m}{p}\binom{N-m+p}{p}$, distinct from the physical J_s convention.
- Oriented Wick marks: 765 geometry cases, 6120 exact coefficient comparisons, 27 pair-site rows, 446 legal two-slot cases, and 5760 finite trace comparisons. Selected finite words include a recorded random seed; this is not exhaustive over all index sets. The invalid raw nonadjacent shortcut is retained as an expected failure, with its hypothesis violation identified.
- Whole-field sources: 6408 exact finite-CAR scalar coefficients, with internal contractions allowed. Small systems use every index assignment; larger cases use specified deterministic assignments. All 294 tested no-bridge diagrams satisfy (34); 80 have a nonzero connected coefficient. These finite checks accompany the local proof, not an extrapolation from a handful of large-N points.
- Charge background: 81 exact (N, p) systems and 1643 sectors; 12 full binomial sums at 70 and 110 decimal digits, and 72 central points. No charge cutoff is used in these sums. At $p = 128$, the variance relative error is still about 1.73% for $\lambda = 1/4$ and 8.65% for $\lambda = 1$. Precision agreement is not an interval certificate or a substitute for the analytic limit.
- Closed geometry: 2368 diagrams and 18944 source/G coefficient comparisons, including 724 diagrams with internal pairs and 277 mixed cases with internal D pairs. Pure-Wick checks give 6576 further comparisons. Negative coefficients are retained; the three distinct G_2 norm subtractions are recorded separately.

The immediate next task is the full joint charge/energy regression of G_n and K_n, followed by the charged residual two-energy measures and their mixed measures with the neutral descendants. In particular, the known nonzero ν_{A_0,B_2} must survive. The uniform tails (16)–(19) then provide the finite-memory-time passage. The complete joint memory kernel, the de Sitter bulk dictionary, and finite-G gravitational dynamics are not claimed here. Work remains local; this stage makes no Git commit, push, or external issue reply.

References

[1] M. Berkooz, V. Narovlansky, and H. Raj, *Complex Sachdev-Ye-Kitaev model in the double scaling limit*, arXiv:2006.13983, Secs. 3.2–3.3 and Eq. (4.3). Original fermions have anticommutator 2; our finite checks use standard CAR.

[2] A. Day and Chengheng, *Adjacent-charge transfer and Gaussian OU propagation*, local project stage `susskind-memory-charged-transfer` (2026).

[3] A. Day and Chengheng, *Uniform finite-memory bounds*, local project stage `susskind-memory-uniform-bound` (2026).

[4] A. Day and Chengheng, *Two-component finite-memory reduction*, local project stage `susskind-memory-two-component` (2026).

[5] A. Day and Chengheng, *A constructive neutral memory kernel*, local project stage `susskind-memory-wick-descendants` (2026).